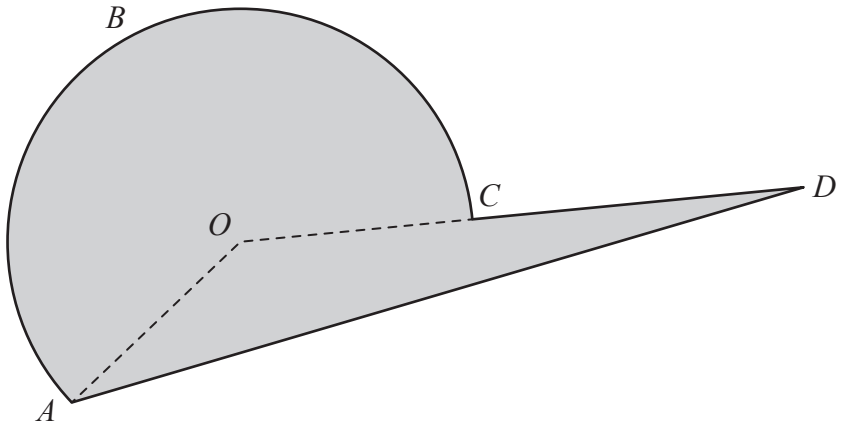
16 Here is a shaded shape *ABCD*.



The shape is made from a triangle and a sector of a circle, centre *O* and radius 6 cm.
OCD is a straight line.

AD = 14 cm
Angle *AOD* = 140°
Angle *OAD* = 24°

Calculate the perimeter of the shape.
Give your answer correct to 3 significant figures.

Major angle = 360° − 140° = 220°

Whole circumference = $2\pi \times 6 = 12\pi$ cm

	Degrees	Arc (cm)	
×11/18	360	12π	×11/18
	220	$\frac{22\pi}{3}$	

1 mark
Scale the full circle to the major arc.

Sine rule: $\frac{OD}{\sin 24°} = \frac{14}{\sin 140°}$

1 mark
Correct sine-rule relationship.

$OD = 14 \sin 24° \div \sin 140° = 8.858778416...$

1 mark
Find OD.

Perimeter = 23.03834613... + 14 + 8.858778416... − 6

1 mark
Add the major arc, AD and CD = OD − 6.

1 mark
Correct perimeter (cm).

39.9 .....cm

(Total for Question 16 is 5 marks)